

IMDB Movie Analysis

- In this data set there more than 5000columns and have duplicates
- Had the blanks in the data set

Data cleaning:

Cleaning the data:

- This help me a lot
- Clening the duplicates and blanks

My analysis:

- The big movie director as only one movie and they are top 10%
- Thre biggest movie avatar as highest movie profit margin & gross also We analyzed how

Story

movie budgets impact financial success by comparing budgets and gross earnings. While higher budgets generally led to higher earnings, some low-budget movies earned huge profits, showing that spending more doesn't always guarantee success. By calculating profit and profit margin, we identified movies that were most efficient in turning investment into earnings. The scatter plot confirmed a positive trend between budget and gross, but also highlighted outliers — low-budget blockbusters that achieved remarkable returns. Overall, smart investment and audience appeal often matter more than just a big budget.

We analyzed factors affecting movie ratings, including budget, genre, and director. Higher-budget movies often receive better ratings, likely because they can afford higher production quality, which improves the **viewer** experience. Positive experiences lead to better reviews, which in turn influence more people to watch the movie, boosting its popularity.

Key Insight: While budget helps, director, genre, storytelling, and creativity are also crucial—some low-budget films achieve high ratings due to strong content.

Conclusion: Success depends on a mix of quality, audience experience, and creative execution. Visualizations like scatter plots and bar charts confirmed these patterns, helping producers and marketers make informed decisions.

Tasks

A. Movie Genre Analysis: Analyze the distribution of movie genres and their impact on the IMDB score.

Task: Determine the most common genres of movies in the dataset. Then, for each genre, calculate descriptive statistics (mean, median, mode, range, variance, standard deviation) of the IMDB scores.

Analysis:

- The most common genre of movies is **drama**
- the analysis done in the Excel sheet

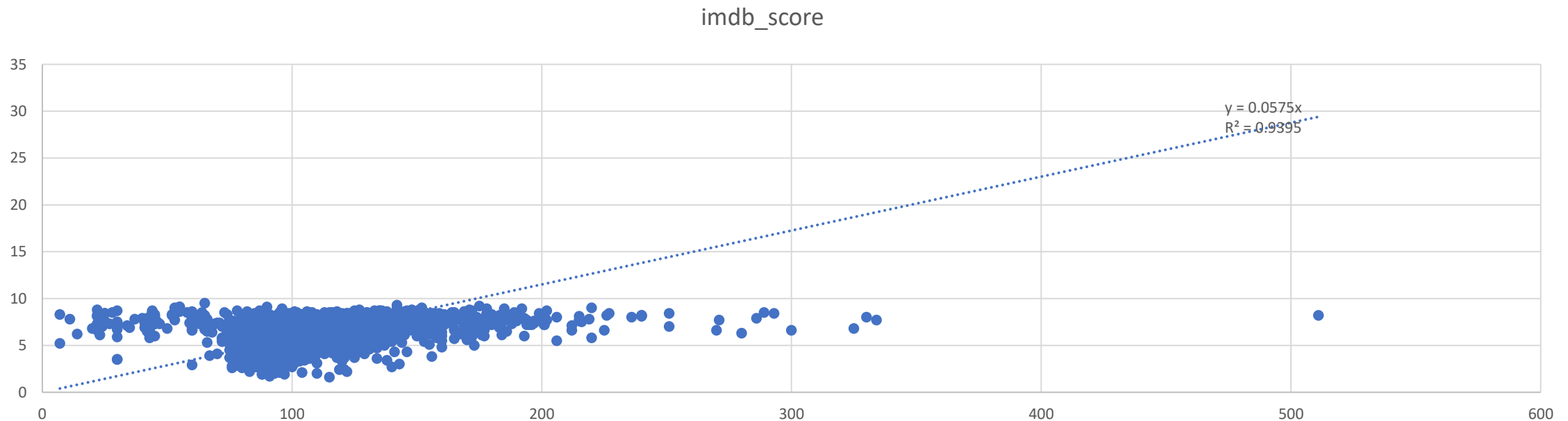
<https://docs.google.com/spreadsheets/d/19gpWPkkX4aHGDEPxQMt-tLvU3gt7T8HX/edit?usp=sharing&ouid=114401290432215454951&rtpof=true&sd=true>

B. Movie Duration Analysis: Analyze the distribution of movie durations and its impact on the IMDB score.

Task: Analyze the distribution of movie durations and identify the relationship between movie duration and IMDB score.

Analysis:

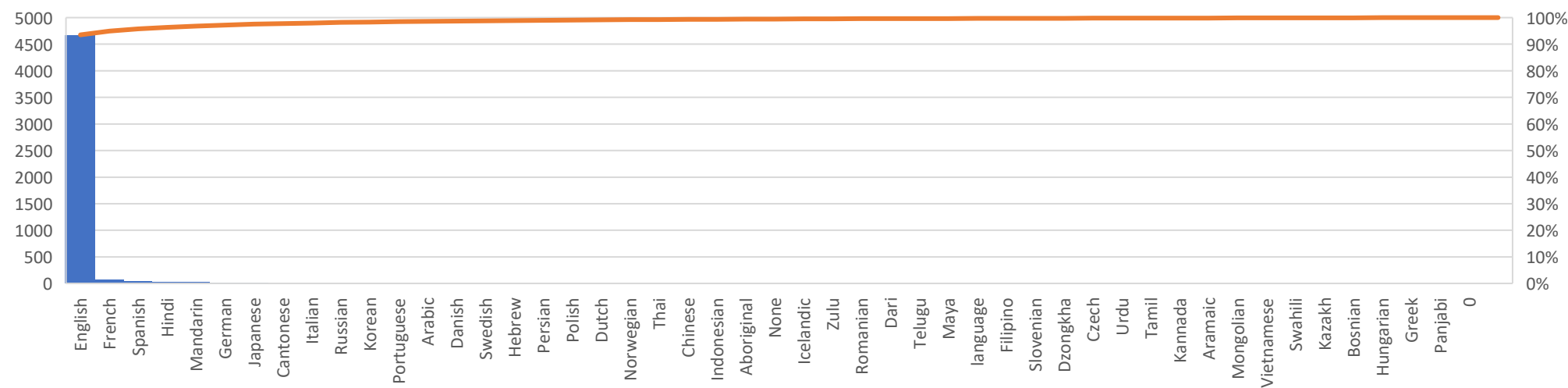
- The average duration have average score
- The is done by vizuazation



Language Analysis: Situation: Examine the distribution of movies based on their language.
Task: Determine the most common languages used in movies and analyze their impact on the IMDB score using descriptive statistics.

- Analysis:
- The most common language is English
 - The analysis is done by common descriptive

Chart Title



Director Analysis: Influence of directors on movie ratings.

Task: Identify the top directors based on their average IMDB score and analyze their contribution to the success of movies using percentile calculations.

Analysis:

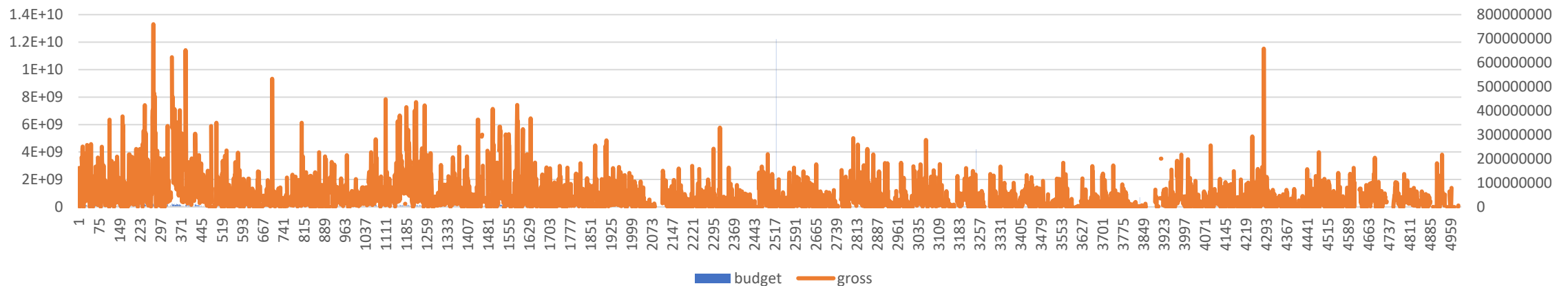
- The top 10% directors had mostly one movie
- <https://docs.google.com/spreadsheets/d/19gpWPkkX4aHGDEPxQMt-tLvU3gt7T8HX/edit?usp=sharing&oid=114401290432215454951&rtpof=true&sd=true>

Budget Analysis: Explore the relationship between movie budgets and their financial success.
Task: Analyze the correlation between movie budgets and gross earnings, and identify the movies with the highest profit margin.

Analysis:

- The top margin movie is avatar
- The corr realation blw them is 0.101033
- <https://docs.google.com/spreadsheets/d/19gpWPkkX4aHGDEPxQMt-tLvU3gt7T8HX/edit?usp=sharing&ouid=114401290432215454951&rtpof=true&sd=true>

Chart Title



Approch:

We explored the IMDB dataset to analyze factors affecting movie success, including **budget, director, genre, language, and ratings**. The project was executed using **descriptive statistics, correlation analysis, and visualization techniques** to uncover relationships between variables. We also used the **Five Whys approach** to dig deeper into patterns, such as why higher-budget movies often receive higher ratings.

Tech stack:

Microsoft Excel 2023 – for data cleaning, calculations, and analysis using formulas like AVERAGEIF, COUNTIF, CORREL, PERCENTILE, and RANK.

Excel Charts – for scatter plots, bar charts, and trendlines to visualize patterns in ratings, budgets, and profits.

insights

- Higher-budget movies often achieve better ratings, mainly due to better production quality and enhanced viewer experience.
- Some low-budget movies also perform well, highlighting the importance of **storytelling, creativity, and director influence**.
- Profit and profit margin analyses revealed that movies with moderate budgets can yield the highest returns, emphasizing **smart investment over spending more**.
- Patterns across genres and languages showed that audience preferences significantly affect ratings.